

Statistical Analysis Plan: ADNI MCI Conversion Prediction

PHB 243b Project 2

[Your Name Here]

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1 Introduction

1.1 Background and Rationale

[Your background text here]

1.2 Research Questions

Primary Research Question:

[Your primary question here]

Secondary Questions (if any):

1. Which predictor categories (sociodemographic, clinical, neuropsychological) contribute most to prediction accuracy?
2. Does prediction performance differ between EMCI and LMCI subgroups?

1.3 Hypotheses

Primary Hypothesis:

[Your hypothesis here, e.g., “Baseline ADAS-Cog-13, CDR-SB, and RAVLT forgetting scores will significantly predict MCI-to-AD conversion...”]

Expected Performance:

Based on Grassi et al. (2019), we expect AUROC in the range of 0.75-0.85.

2 Data Description

2.1 Data Source

Data for this analysis come from the Alzheimer’s Disease Neuroimaging Initiative (ADNI), accessed via the ADNIMERGE2 R package (Donohue et al., 2026). ADNI is a longitudinal multicenter study designed to develop clinical, imaging, genetic, and biochemical biomarkers for early detection and tracking of Alzheimer’s disease.

Data access: adni.loni.usc.edu

Data download date: [Your download date]

Citation: Donohue MC, et al. (2026). Alzheimer’s clinical research data via R packages: The alzverse. *Alzheimer’s Dement*, 22:e71152.

2.2 Study Population

Inclusion Criteria:

1. Baseline diagnosis of Mild Cognitive Impairment (EMCI or LMCI)
2. At least 3 years of follow-up OR documented conversion to AD
3. Complete data on primary predictor variables

Exclusion Criteria:

1. Baseline diagnosis of CN or Dementia
2. No follow-up visits after baseline
3. Missing data on >50% of predictor variables

2.3 Sample Size

Expected Sample Size:

Group	N (expected)	Percentage
Stable MCI (sMCI)	~350	~64%
Converters (cAD)	~200	~36%
Total	~550	100%

Based on Grassi et al. (2019) cohort selection criteria.

3 Variables

3.1 Primary Outcome

Attribute	Description
Variable name	converter
Definition	Conversion from MCI to AD dementia
Type	Binary (0 = Stable, 1 = Converter)
Operationalization	Any follow-up diagnosis of “Dementia” within 3 years
Reference	Grassi et al. (2019)

3.2 Primary Predictors

3.2.1 Sociodemographic Variables

Variable	Definition	Type	Rationale
AGE	Age at baseline (years)	Continuous	Age is primary AD risk factor
PTEDUCAT	Years of education	Continuous	Cognitive reserve proxy
PTGENDER	Sex	Binary	Sex differences in AD risk
PTMARRY	Marital status	Categorical	Social support proxy

3.2.2 Clinical Scales

Variable	Definition	Type	Rationale
CDRSB	CDR Sum of Boxes (0-18)	Continuous	Clinical staging
FAQ	Functional Activities Questionnaire (0-30)	Continuous	Functional impairment

3.2.3 Neuropsychological Tests

Variable	Definition	Type	Rationale
MMSE	Mini-Mental State Exam (0-30)	Continuous	Global cognition
ADAS13	ADAS-Cog 13-item (0-85)	Continuous	AD-specific cognition
RAVLT_immediate	RAVLT sum of trials 1-5	Continuous	Verbal learning
RAVLT_learning	RAVLT trial 5 - trial 1	Continuous	Learning rate
RAVLT_forgetting	RAVLT trial 5 - delayed	Continuous	Memory retention
LDELTOTAL	Logical Memory delayed recall	Continuous	Episodic memory
TRABSCOR	Trail Making Test B (seconds)	Continuous	Executive function

3.2.4 MCI Subtype

Variable	Definition	Type	Rationale
DX_BL	Baseline MCI subtype	Binary	EMCI vs LMCI

3.3 Excluded Variables

The following variable categories are intentionally **excluded** to maximize clinical translatability (Grassi et al., 2019):

- **Imaging:** MRI volumes, PET SUVRs
- **Genetic:** APOE genotype
- **Biomarkers:** CSF Abeta, tau, p-tau

Rationale: These measures require specialized equipment and expertise not available in most community clinical settings.

4 Statistical Methods

4.1 Descriptive Statistics

Continuous variables: Mean (SD), median (IQR), range

Categorical variables: Frequencies, percentages

Stratification: By conversion status (sMCI vs cAD)

Comparison tests:

- Continuous: t-test or Wilcoxon rank-sum
- Categorical: Chi-square or Fisher's exact

4.2 Missing Data

Strategy: [Complete case analysis / Multiple imputation / Specify]

Justification: [Your justification here]

4.3 Machine Learning Methods

4.3.1 Algorithms

We will implement at least three classification algorithms:

Algorithm	R Package	Key Hyperparameters
Logistic Regression	<code>glm</code>	None
Random Forest	<code>randomForest</code>	<code>ntree</code> , <code>mtry</code>
Gradient Boosting [Optional: Additional]	<code>xgboost</code>	<code>max_depth</code> , <code>eta</code> , <code>nrounds</code>

4.3.2 Cross-Validation Strategy

1. **Train/Test Split:** 70% training, 30% testing
2. **Cross-Validation:** 5-fold CV within training set
3. **Site Independence:** [If feasible, split by ADNI site]

4.3.3 Hyperparameter Tuning

Algorithm	Tuning Method	Grid/Range
Random Forest	Grid search	<code>mtry</code> : 2-8, <code>ntree</code> : 100-500
XGBoost	Grid search	<code>max_depth</code> : 3-6, <code>eta</code> : 0.01-0.3

4.4 Evaluation Metrics

Metric	Definition	Target
AUROC	Area under ROC curve	> 0.75
Sensitivity	$TP / (TP + FN)$	> 0.70
Specificity	$TN / (TN + FP)$	> 0.70
PPV	$TP / (TP + FP)$	Report
NPV	$TN / (TN + FN)$	Report

Threshold Selection: Youden’s J statistic (maximize sensitivity + specificity)

4.5 Feature Importance

- **Random Forest:** Mean decrease in Gini / Permutation importance
- **Logistic Regression:** Standardized coefficients
- **XGBoost:** Gain-based importance

5 Planned Results

5.1 Table 1: Baseline Characteristics by Conversion Status

Characteristic	Stable MCI	Converters	p-value
N			
Age, mean (SD)			
Education, mean (SD)			
Female, n (%)			
EMCI, n (%)			
MMSE, mean (SD)			
ADAS-Cog-13, mean (SD)			
CDR-SB, mean (SD)			
FAQ, mean (SD)			
RAVLT immediate, mean (SD)			
RAVLT forgetting, mean (SD)			
LDELTOTAL, mean (SD)			
TMT-B, mean (SD)			

5.2 Table 2: Model Performance Comparison

Metric	Logistic Reg	Random Forest	XGBoost
AUROC (95% CI)			
Sensitivity			
Specificity			
PPV			
NPV			
Accuracy			

5.3 Table 3: Feature Importance Rankings

Rank	Logistic Reg	Random Forest	XGBoost
1			
2			
3			
4			
5			

5.4 Planned Figures

5.4.1 Figure 1: Cohort Flow Diagram

- **Type:** CONSORT-style flow diagram
- **Purpose:** Show sample selection and exclusions

5.4.2 Figure 2: ROC Curves

- **Type:** Overlaid ROC curves for all models
- **Purpose:** Compare discrimination across algorithms

5.4.3 Figure 3: Feature Importance Plot

- **Type:** Bar chart or dot plot
- **Purpose:** Visualize top predictors

5.4.4 Figure 4: Calibration Plot

- **Type:** Predicted vs. observed probabilities
- **Purpose:** Assess calibration of best model

6 Sensitivity Analyses

Analysis	Rationale	Expected Impact
EMCI only	Test in early-stage MCI	Lower conversion rate
LMCI only	Test in late-stage MCI	Higher conversion rate
Exclude outliers	Robustness check	Minimal change expected
Alternative imputation	Compare to complete case	Similar results expected
Different CV folds	Check stability	Similar AUROC expected

7 Limitations

7.1 Scope and Generalizability

- Results specific to ADNI population (highly educated, mostly White)
- May not generalize to community clinic populations
- ADNI uses strict diagnostic criteria

7.2 Methodological Limitations

- Retrospective cohort design
- Cannot establish causation
- 3-year prediction horizon may miss later converters
- No external validation cohort

7.3 Data Limitations

- Missing data on some predictor variables
- Potential selection bias in ADNI enrollment
- Measurement variability across sites

8 Timeline

Week	Task	Deliverable
1 (Feb 17-21)	Data preparation, cohort selection	Clean dataset
2 (Feb 24-28)	Feature engineering, initial models	Preliminary results
3 (Mar 2-6)	Model tuning, evaluation, report	Final submission

Due Date: March 3, 2026

9 References

1. Donohue MC, et al. (2026). Alzheimer's clinical research data via R packages: The alzverse. *Alzheimer's Dement*, 22:e71152.
2. Grassi M, et al. (2019). A Clinically-Translatable Machine Learning Algorithm for the Prediction of Alzheimer's Disease Conversion. *Front Neurol*, 10:756.
3. Petersen RC. (2004). Mild cognitive impairment as a diagnostic entity. *J Intern Med*, 256(3):183-194.
4. Rye I, et al. (2022). Predicting conversion to Alzheimer's disease in individuals with Mild Cognitive Impairment. *Sci Rep*, 12:15566.

10 Appendix: Code Framework

```
# =====  
# Data Preparation  
# =====  
library(ADNIMERGE2)  
library(tidyverse)  
library(caret)  
  
# Load data
```

```

data(ADSL)
data(ADQS)

# Select MCI subjects at baseline
mci_baseline <- ADSL |>
  filter(DX_BL %in% c("EMCI", "LMCI"))

# Define conversion status
# [Your code here]

# =====
# Feature Engineering
# =====
# Select predictor variables
predictors <- c("AGE", "PTEDUCAT", "PTGENDER", "CDRSB", "FAQ",
               "MMSE", "ADAS13", "RAVLT_immediate", "RAVLT_forgetting",
               "LDELTOTAL", "TRABSCOR")

# =====
# Model Training
# =====
# Train/test split
set.seed(243)
train_idx <- createDataPartition(mci_data$converter, p = 0.7, list = FALSE)

# Cross-validation setup
ctrl <- trainControl(method = "cv", number = 5,
                     classProbs = TRUE, summaryFunction = twoClassSummary)

# =====
# Model Evaluation
# =====
# ROC curves
library(pROC)
# [Your code here]

```